OCR Model Comparison

PaddleOCR, GLM-OCR and Tesseract OCR

This document shows the same input image, the extracted text, and the saved response from three OCR models. The examples use real project results.

Model	Returns	Runs through
PaddleOCR	Structured Aadhaar fields	Python / CPU
GLM-OCR	Raw text and metadata	Local Ollama
Tesseract OCR	Raw text and line confidence	Python / CPU

Input image

File: apeksh.webp | Size: 768 × 865 pixels | Content: front and back of an Aadhaar card.



The same image is used for all three models.

Reading the results: OCR errors are kept as returned. PaddleOCR also uses an Aadhaar field parser. GLM-OCR and Tesseract return raw text with metadata. This single example is not an accuracy benchmark.

1. PaddleOCR

Model: PP-OCRv5_server_det + en_PP-OCRv5_mobile_rec. A targeted devanagari_PP-OCRv5_mobile_rec pass reads the Hindi name. Package: PaddleOCR 3.7.0.

Input: apeksh.webp (the image shown above).

```
python main.py apeksh.webp --engine paddleocr --output output\comparison
```

Output: extracted text

```
HR RT TE
UNIQUE IDENTIFICATION AUTHORITY OF INDIA
3TET
Download Date: 05/12/2019
YaT:
D/O/
2HT4RH, 3T 1SR RTE, 2T ST.3H
2i, $I1RGS, 834001
Address:
D/O: Sanjiv Kumar Mishra, gopal ganj / upper
bazar, near shani mandir, upper bazar ranchi,
Ranchi G.P.O., Ranchi, Jharkhand, 834001
7458 6035 2502
WWW
P.O. Box No.1947,
idai gov.in
Rengaluru-560 001
AIRD
HROR
GOVERNMENT OF INDIA
mat
3FT
Apeksha Mishra
30H
/D23/07/2003
/FEMALE
7458 6035 2502
T3TETR
A
46419
Goo0oj pa
Download Date: 05/12/2019
अपेक्षा मिश्रा
```

PaddleOCR: JSON response

This is the complete structured response saved by main.py. Dates use YYYY-MM-DD. Empty strings mean that a value was not extracted.


```
{
  "document_type": "Aadhaar",
  "name": "Apeksha Mishra",
  "name_hindi": "अपेक्षा मिश्रा",
  "date_of_birth": "2003-07-23",
  "gender": "Female",
  "aadhaar_number": "745860352502",
  "virtual_id": "",
  "issue_date": "",
  "download_date": "2019-12-05",
  "address": {
    "care_of": "Sanjiv Kumar Mishra",
    "house_number": "",
    "locality": "Gopal Ganj / Upper Bazar, Near Shani Mandir, Upper Bazar Ranchi",
    "city": "Ranchi",
    "district": "Ranchi",
    "state": "Jharkhand",
    "pin_code": "834001",
    "country": "India"
  },
  "issuing_authority": "UIDAI",
  "qr_code_present": true
}
```


The raw OCR text contains recognition errors. The additional field parser and Hindi-name pass produce the structured fields shown here.

2. GLM-OCR

Model: glm-ocr:q8_0. Runtime: local Ollama. Module: glm_ocr.py.

Input: apeksh.webp (the image shown above).

```
python main.py apeksh.webp --engine glm
```

Output: extracted text

```
आयात
भारतीय विश्वास पहचान प्राधिकरण
UNIQUE IDENTIFICATION AUTHORITY OF INDIA
पता:
D/O: संजीव कमार मिश्रा, गौपालल गौपज / अपर बाजार,
शाखी मठिर के पासस, अपर बाजार राखी, राखी जी.पी.आ
राखी, दुरार्थड, 834001
Address:
D/O: Sanjiv Kumar Mishra, gopal ganj / upper
bazar, near shani mandir, upper bazar ranchi,
Ranchi G.P.O., Ranchi, Jharkhand, 834001
Download Date: 05/12/2019
7458 6035 2502
www
uidai.gov.in
P.O. Box No.1947,
Bengaluru-560 001
भारत सरकार
GOVERNMENT OF INDIA
अपेद्धा मिश्रा
Apeksha Mishra
जन्म तिनिय / DOB: 23/07/2003
मिश्रा / FEMALE
7458 6035 2502
मेरा आयात, मेरी पहचान
```

JSON response: excerpt

The text field is shown in full above. This excerpt shows the model and generation metadata.


```
{
  "engine": "glm",
  "model": "glm-ocr:q8_0",
  "truncated": false,
  "usage": {
    "prompt_tokens": 849,
    "generated_tokens": 445,
    "total_duration_ns": 37862779000
  }
}
```


Observation: the English name, DOB and address were read. Some Hindi words were misread. The current GLM route does not create separate Aadhaar fields.

3. Tesseract OCR

Model: Tesseract 5.4.0 with English language data (eng). Python wrapper: pytesseract 0.3.13.

Input: apeksh.webp (the image shown above).

```
python main.py apeksh.webp --engine tesseract --output output\comparison
```

Output: extracted text

```
= Gediatartrensa wif _
‘BN. UNIQUE)IDENTIFICATION/AUTHORITY OF INDIA
War.
D/O: Satta Hare FAT, HTT TT / HTT ATA
alfa Afex & Oe, rT ah aa ah aH
If, ARGS, 834001
® Address:
& D/O: Sanjiv Ku
bazar, near shani ma
ancni G.P.O., Ranchi, J
@42q peojuMog
mar Mishra, gopal ganj / upper
indir, upper bazar ranchi,
warknand, 634001
SO
R
602/24,
P.0. Box No.1947,
Renaaluru-560 001
Toa Fat
|
. Apeksha Mishra
wem fete / DOB: 23/07/2003 meres
Ar Tert
,| after / FEMALE 5 fe
ae ae eae
i eo
7458 6035 2502
f
\
```

JSON response: excerpt

The full text is shown above. This excerpt includes the first line and its confidence score. The complete saved response contains all line records.


```
{
  "engine": "tesseract",
  "language": "eng",
  "lines": [
    {
      "text": "= Gediatartrensa wif _",
      "confidence": 0.39749999999999996
    }
  ]
}
```


Observation: the name, DOB and Aadhaar number appear in the output. Some address lines are split or misread. English-only language data does not reliably read Hindi.